Experiment 6: Payroll Processing System

Program (Python 3)

```
employees = [
    ("Suresh", 5, 12, 40000),
    ("Lakshmi", 3, 8, 30000),
    ("Vignesh", 10, 15, 65000),
]
print(f"{'Name':10}{'ExpHere':8}{'TotalExp':9}{'Basic':8}{'HRA':8}{'DA':8}{'TA':7}{'Gross':9}{'Deduct':8}{'Net'}")
for name, exp_here, exp_all, basic in employees:
    hra = 0.20 * basic
    da = 0.50 * basic
    ta = 3000
    gross = basic + hra + da + ta
    pf = 0.12 * basic
    tax = 0.05 * gross if gross > 50000 else 0
    ded = pf + tax
    net = gross - ded
    print(f"{name:10}{exp_here:<8}{exp_all:<9}{basic:<8.0f}{hra:<8.0f}{da:<8.0f}{ta:<7}{gross:<9.0f}{ded:<8.0f}{net:.0f}")
```

Output

```
Name      ExpHere TotalExp Basic   HRA     DA      TA     Gross    Deduct  Net
Suresh    5       12       40000   8000    20000   3000   71000    8350    62650
Lakshmi   3       8        30000   6000    15000   3000   54000    6300    47700
Vignesh   10      15       65000   13000   32500   3000   113500   13475   100025
```

Reference output generated by running the program above (Python 3.13 / gcc). Not a screenshot of an online compiler.

Experiment 7: Students Attendance System

Program (Python 3)

```
total_classes = 50
students = [
    ("Arun", "CSA15", 45, 5, 0),
    ("Divya", "CSA15", 38, 12, 4),     # name, course, present, absent, OD days
    ("Karthik", "CSA15", 30, 20, 0),
    ("Meena", "CSA15", 42, 8, 2),
]
def grade(p):
    return "O" if p >= 95 else "A" if p >= 85 else "B" if p >= 75 else "C" if p >= 65 else "Detained"
print(f"{'Name':9}{'Course':8}{'Present':8}{'Absent':7}{'OD':4}{'Attend%':9}{'Grade':9}Remark")
for n, c, p, a, od in students:
    pct = (p + od) / total_classes * 100
    remark = "Eligible" if pct >= 75 else "Short attendance"
    print(f"{n:9}{c:8}{p:<8}{a:<7}{od:<4}{pct:<9.1f}{grade(pct):9}{remark}")
```

Output

```
Name     Course  Present Absent OD  Attend%  Grade    Remark
Arun     CSA15   45      5      0   90.0     A        Eligible
Divya    CSA15   38      12     4   84.0     B        Eligible
Karthik  CSA15   30      20     0   60.0     Detained Short attendance
Meena    CSA15   42      8      2   88.0     A        Eligible
```

Reference output generated by running the program above (Python 3.13 / gcc). Not a screenshot of an online compiler.

Experiment 8: Student Counselling System

Program (Python 3)

```
students = [
    {"name": "Ajay", "age": 18, "address": "Chennai", "phone": "9876543210",
     "subjects": ["Physics", "Chemistry", "Maths"], "year": 2026, "pct": 92.5, "interest": ""},
    {"name": "Bhavana", "age": 18, "address": "Madurai", "phone": "9123456780",
     "subjects": ["Biology", "Chemistry", "Physics"], "year": 2026, "pct": 85.0, "interest": ""},
    {"name": "Chandru", "age": 19, "address": "Salem", "phone": "9988776655",
     "subjects": ["Commerce", "Accounts", "Maths"], "year": 2025, "pct": 70.0, "interest": ""},
]
def counsel(s):
    subj = " ".join(s["subjects"])
    if "Maths" in subj and "Physics" in subj: dom = "Computer Science / Engineering"
    elif "Biology" in subj: dom = "Medicine / Biotechnology / Pharmacy"
    else: dom = "Management / Commerce / Finance"
    s["interest"] = dom
    return dom
for s in students:
    d = counsel(s)
    print("\n=== STUDENT COUNSELLING RECORD ===")
    print("Name:", s["name"], "| Age:", s["age"])
    print("Address:", s["address"], "| Phone:", s["phone"])
    print("Subjects:", ", ".join(s["subjects"]), "| Year of passing:", s["year"], "| Marks %:", s["pct"])
    print("Recommended Domain:", d)
    print("Scholarship eligible:", "Yes" if s["pct"] >= 90 else "No")
```

Output

```
=== STUDENT COUNSELLING RECORD ===
Name: Ajay | Age: 18
Address: Chennai | Phone: 9876543210
Subjects: Physics, Chemistry, Maths | Year of passing: 2026 | Marks %: 92.5
Recommended Domain: Computer Science / Engineering
Scholarship eligible: Yes

=== STUDENT COUNSELLING RECORD ===
Name: Bhavana | Age: 18
Address: Madurai | Phone: 9123456780
Subjects: Biology, Chemistry, Physics | Year of passing: 2026 | Marks %: 85.0
Recommended Domain: Medicine / Biotechnology / Pharmacy
Scholarship eligible: No

=== STUDENT COUNSELLING RECORD ===
Name: Chandru | Age: 19
Address: Salem | Phone: 9988776655
Subjects: Commerce, Accounts, Maths | Year of passing: 2025 | Marks %: 70.0
Recommended Domain: Management / Commerce / Finance
Scholarship eligible: No
```

Reference output generated by running the program above (Python 3.13 / gcc). Not a screenshot of an online compiler.

Experiment 9: Hospital Information System (SIMATS)

Program (Python 3)

```
doctors = {"Fever": "Dr. Ramesh (General Medicine)", "Fracture": "Dr. Priya (Orthopedics)",
           "Heart": "Dr. Anand (Cardiology)", "Skin": "Dr. Sana (Dermatology)"}
patients = []
def register(name, age, addr, phone, email, att_name, att_phone, disease):
    pid = "P" + str(len(patients) + 1).zfill(3)
    patients.append({"id": pid, "name": name, "age": age, "addr": addr, "phone": phone, "email": email,
                     "att": att_name, "attph": att_phone, "disease": disease,
                     "doctor": doctors.get(disease, "Dr. Kumar (General)")})
    return pid
register("Gopal", 45, "Chennai", "9000011111", "gopal@mail.com", "Sita", "9000022222", "Heart")
register("Hema", 30, "Poonamallee", "9000033333", "hema@mail.com", "Raj", "9000044444", "Fever")
register("Imran", 22, "Avadi", "9000055555", "imran@mail.com", "Salma", "9000066666", "Fracture")
print("=== SIMATS HOSPITAL - PATIENT RECORDS ===")
for p in patients:
    print(f"\nID: {p['id']} | Name: {p['name']} | Age: {p['age']}")
    print(f"Address: {p['addr']} | Phone: {p['phone']} | Email: {p['email']}")
    print(f"Attender: {p['att']} ({p['attph']})")
    print(f"Disease: {p['disease']} | Doctor: {p['doctor']}")
```

Output

```
=== SIMATS HOSPITAL - PATIENT RECORDS ===

ID: P001 | Name: Gopal | Age: 45
Address: Chennai | Phone: 9000011111 | Email: gopal@mail.com
Attender: Sita (9000022222)
Disease: Heart | Doctor: Dr. Anand (Cardiology)

ID: P002 | Name: Hema | Age: 30
Address: Poonamallee | Phone: 9000033333 | Email: hema@mail.com
Attender: Raj (9000044444)
Disease: Fever | Doctor: Dr. Ramesh (General Medicine)

ID: P003 | Name: Imran | Age: 22
Address: Avadi | Phone: 9000055555 | Email: imran@mail.com
Attender: Salma (9000066666)
Disease: Fracture | Doctor: Dr. Priya (Orthopedics)
```

Reference output generated by running the program above (Python 3.13 / gcc). Not a screenshot of an online compiler.

Experiment 10: C Program - Student Information

Program (C)

```c
#include <stdio.h>
struct Student {
    char name[30]; int reg; char address[40]; char phone[15]; int age;
    char course[20]; int marks[3]; int attended, total;
};
int main() {
    struct Student s[2] = {
        {"Arun", 192211001, "Chennai", "9876543210", 19, "B.Tech CSE", {88, 92, 79}, 45, 50},
        {"Divya", 192211002, "Madurai", "9123456780", 18, "B.Tech IT", {95, 89, 91}, 38, 50}
    };
    for (int i = 0; i < 2; i++) {
        int sum = s[i].marks[0] + s[i].marks[1] + s[i].marks[2];
        float avg = sum / 3.0, att = s[i].attended * 100.0 / s[i].total;
        printf("\n===== STUDENT INFORMATION =====\n");
        printf("Name: %s\nReg No: %d\nAddress: %s\nPhone: %s\nAge: %d\nCourse: %s\n",
               s[i].name, s[i].reg, s[i].address, s[i].phone, s[i].age, s[i].course);
        printf("--- Semester Mark Sheet ---\nSub1:%d Sub2:%d Sub3:%d Total:%d Avg:%.2f\n",
               s[i].marks[0], s[i].marks[1], s[i].marks[2], sum, avg);
        printf("Grade: %c\n", avg >= 90 ? 'O' : avg >= 80 ? 'A' : avg >= 70 ? 'B' : 'C');
        printf("--- Attendance Report ---\nAttendance: %.1f%%\n", att);
        printf("--- Progress Report ---\n%s\n", avg >= 75 ? "Excellent progress" : "Needs improvement");
    }
    return 0;
}
```

Output

```
===== STUDENT INFORMATION =====
Name: Arun
Reg No: 192211001
Address: Chennai
Phone: 9876543210
Age: 19
Course: B.Tech CSE
--- Semester Mark Sheet ---
Sub1:88 Sub2:92 Sub3:79 Total:259 Avg:86.33
Grade: A
--- Attendance Report ---
Attendance: 90.0%
--- Progress Report ---
Excellent progress

===== STUDENT INFORMATION =====
Name: Divya
Reg No: 192211002
Address: Madurai
Phone: 9123456780
Age: 18
Course: B.Tech IT
--- Semester Mark Sheet ---
Sub1:95 Sub2:89 Sub3:91 Total:275 Avg:91.67
Grade: O
--- Attendance Report ---
Attendance: 76.0%
--- Progress Report ---
Excellent progress
```

Reference output generated by running the program above (Python 3.13 / gcc). Not a screenshot of an online compiler.